FINA6229
Machine Learning in Finance

Project 3
Due Date: 12:00 PM, 2 April 2025
Credit Card Approval

Instruction: You are encouraged to work in groups (but not across groups). However, you should independently write up and submit your own assignments through the CUHK Blackboard under the folder: *FINA6229FA>>Course Contents>>Lecture 3>>Project 3*. Demo code (as an image file) is provided under the same folder. In addition to answer the questions below, you are asked to **add some comments right after the "#" whenever it appears in the demo code**, to explain the purpose of the corresponding code.

As an example,

```
#
monthly = pd.read_csv('monthly2020.csv', index_col=0, header=None)
T = monthly.shape[0]
```

you are supposed to submit the code with some comments as follows:

```
# import monthly data using the pandas module
monthly = pd.read_csv('monthly2020.csv', index_col=0, header=None)
T = monthly.shape[0]
```

1. Background

A tech startup Credit.ai is looking to automate the credit card approval process using machine learning. By automating this task, thousand hours of labor costs can be saved. Credit.ai's first client, NCBHK (Nobel Commercial Bank of Hong Kong), handed over some data it has collected on credit card approvals.

Fortunately, Credit.ai has done all the data preparations and has handed over you a .csv file with all the features and labels. It is your job to implement a machine learning algorithm that best mimics a human being using this data. You will be evaluated based on both the true positive and true negative rates of your predictions from a separate set of testing data.

2. The Data

The training data set is *train_data.csv*. Table 1 defines all the features in the data set, and Table 2 gives the definitions for each categorical feature's categories. You are free to use any machine learning methods to train the model.

Table 1: Definition of Variables

Features	Definitions
DEBT	Reported total debt
YRS_IN_RESTDENT	Number of years at current home address
AGE	Reported age
DTI	Debt to income ratio
NUM_PREV_APP	Number of previous submitted applications
OCCUPATION	Reported occupation
PRDVIDED SIN	Whether or not applicant provided a SIN
MARRIAGE	Reported a marriage status
INCOME	Reported income bracket
EDUCATION	Reported highest completed education status
CREDIT_PROFILE	Formed a credit profile matching submitted application
APPROVAL STATUS	Data's label, whether or not applicant is approved

Table 2: Definitions of Categorical Variables

OCCUPATION	
1	N/A
Others	Represents an occupation

PROVIDED_SIN	
0	No SIN Provided
1	Provided a SIN

MARRIAGE	
1	Married
2	Single
3	Other

INCOME	
1	Less than $30,000
2	$30,000–$40,000
3	$40,000–$50,000
4	$50,000–$60,000
5	$60,000–$70,000
6	$70,000–$80,000
7	$80,000–$90,000
8	$90,000–$100,000
9	$100,000–$110,000
10	$110,000–$120,000
11	$120,000–$130,000
12	$130,000–$140,000
13	$140,000–$150,000
14	More than $150,000

EDUCATION	
1	Other
2	High School
3	Associate
4	Bachelor
5	Master
6	PhD
7	Trades

CREDIT_PROFILE	
0	Not Found
1	Found

APPROVAL_STATUS	
0	Not Approved
1	Approved

3. Test the Model

The testing data set is *test_data.csv*. Apply the trained model to evaluate the out-of-sample performance using the testing data.

4. Try a Different Machine Learning Algorithm (optional question with 2 bonus points)

In the demo code, we use the logistic regression to train the model. Please try a different classification method you learned from the class to analyze the credit card data (both **train and test** the model).

5. Please submit this assignment electronically through the CUHK Blackboard as **a single Jupyter Notebook (*.ipynb) with all the outputs and the answers** to the questions above and the code outputs. **Please name the submitted file with project number, your full name, student ID, and group number as**:

ProjectNumber_StudentName_StudentID_GroupNumber.ipynb,

for example: *Project1_GangLi_12345678_Group1.ipynb*.